

APPENDIX

Properties of Si at 300K

Bandgap	1.12 eV
Relative permittivity	11.8
Electron mobility	1350 cm ² /V-s
Hole mobility	480 cm ² /V-s
Effective density of states in the conduction band	$2.8 \times 10^{19} \text{ cm}^{-3}$
Effective density of states in the valence band	$1.04 \times 10^{19} \text{ cm}^{-3}$
Intrinsic carrier concentration	$1.5 \times 10^{10} \text{ cm}^{-3}$

Table of Physical Constants

	Symbol	Value	Units
Planck's constant	h	6.626×10^{-34}	J-s
Speed of light	c	3.0×10^8	m/s
Electronic charge	e (or q)	1.6×10^{-19}	C
Boltzmann's constant	k_B (or k)	1.38×10^{-23}	J/K
Free electron rest mass	m_0	9.1×10^{-31}	kg
Proton rest mass	m_p	1.67×10^{-27}	kg
Permittivity of free space	ϵ_0	8.85×10^{-12}	F/m
Electron-volt	1 eV	1.6×10^{-19}	J
Thermal voltage ($T= 300\text{K}$)	kT/q	0.0259	V